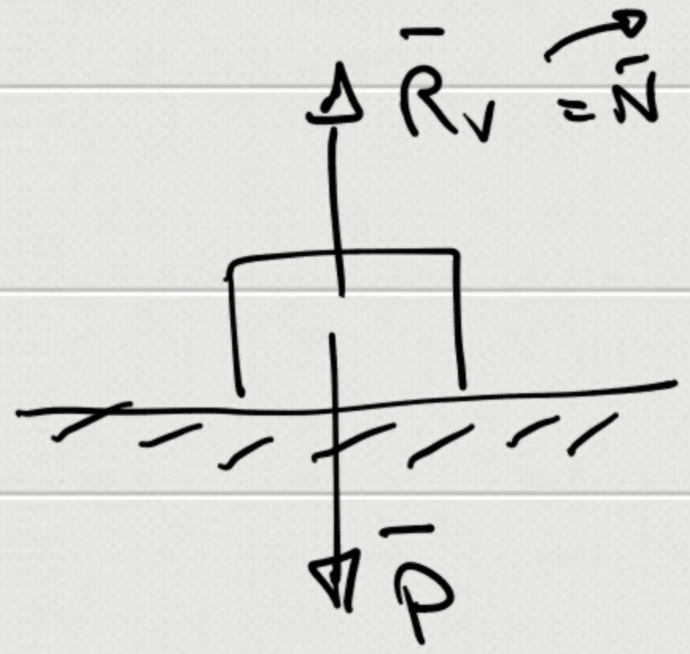
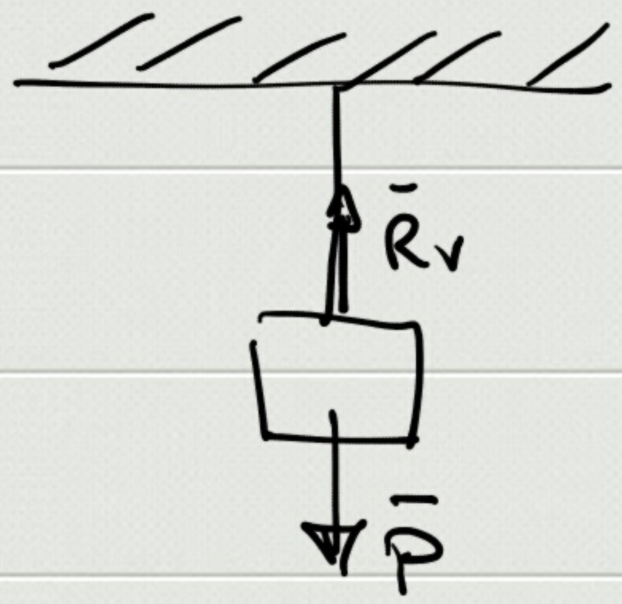
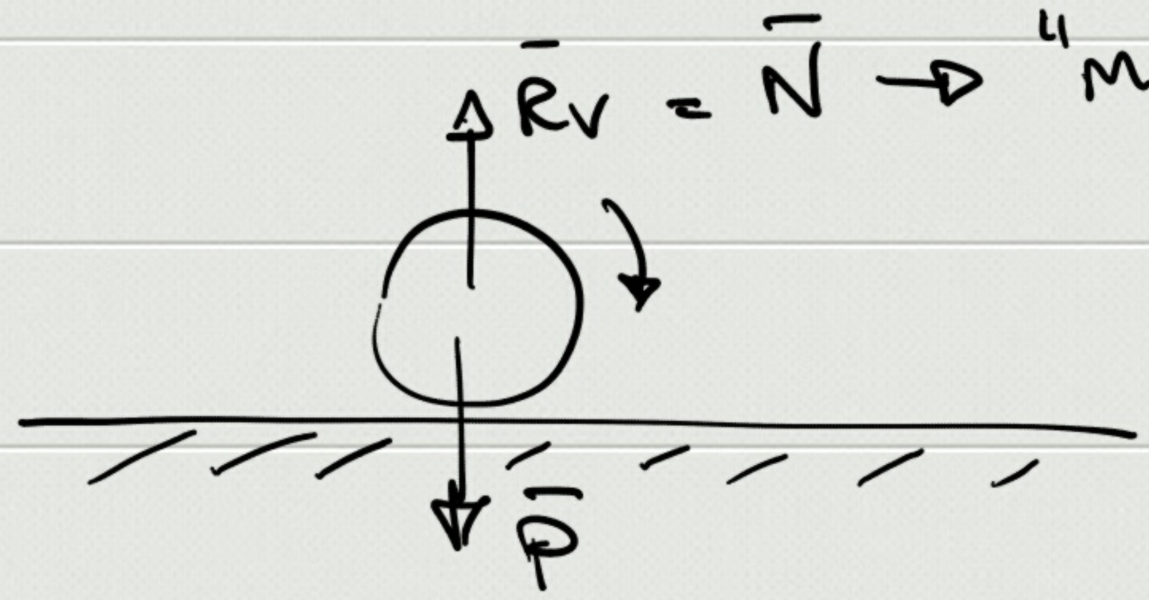
reazione vincolare

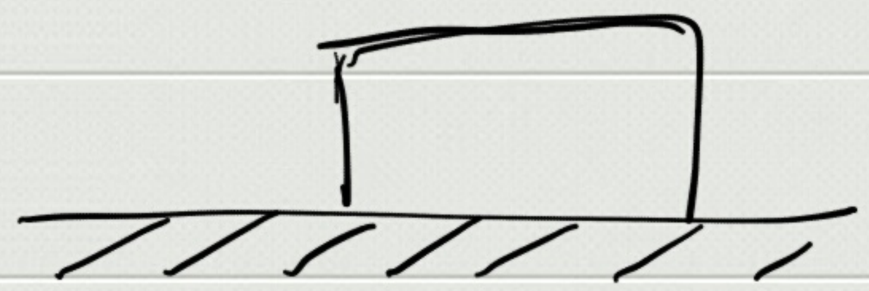


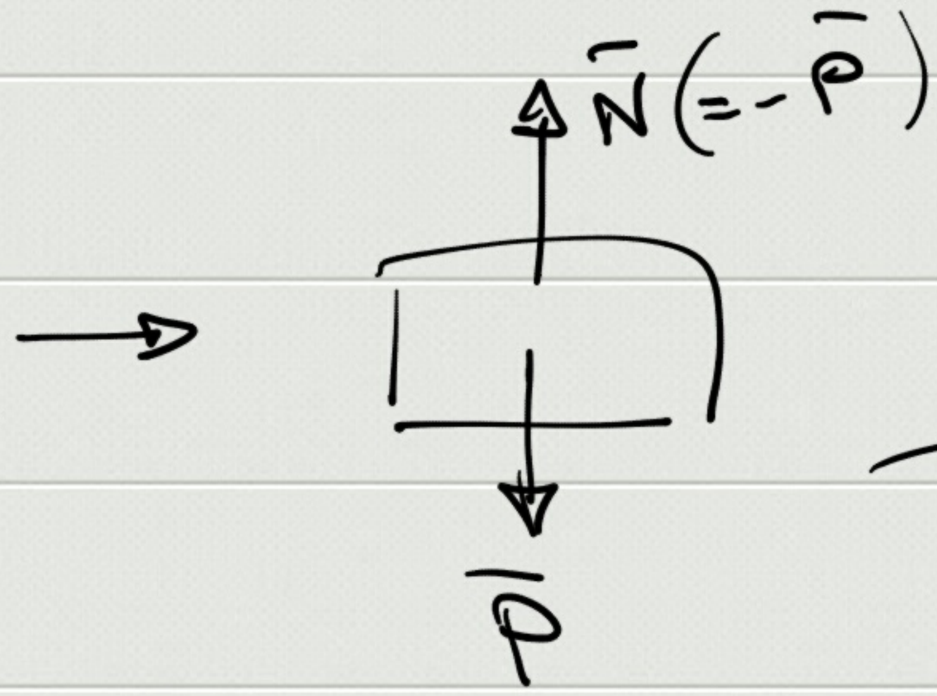
$$\bar{R} = \sum_i \bar{F}_i = m\bar{a}$$

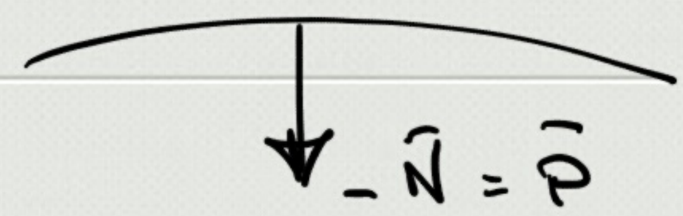
$$\bar{R} = \bar{P} + \bar{R}_v = 0 \Rightarrow \boxed{\bar{R}_v = -\bar{P}}$$

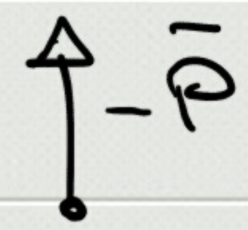


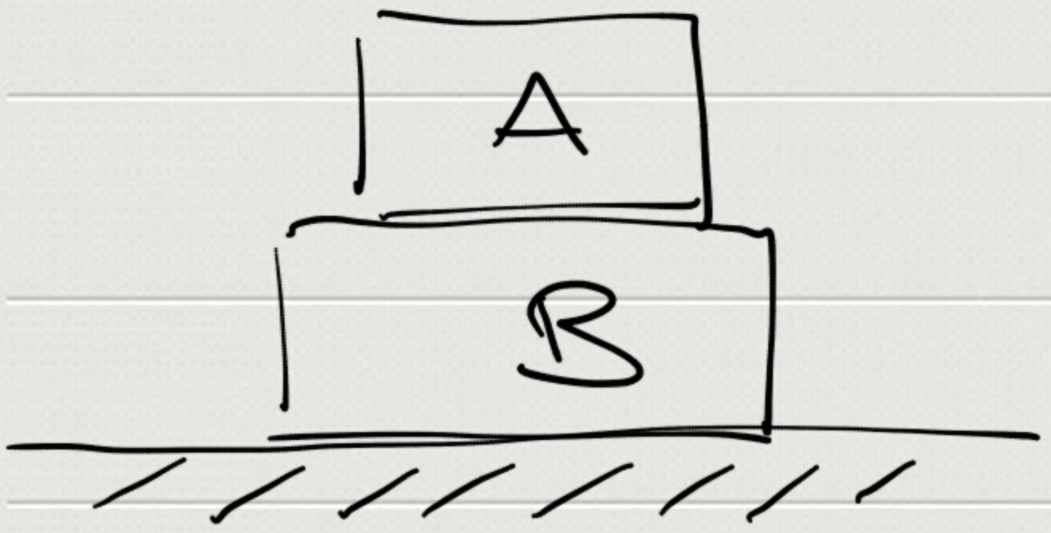


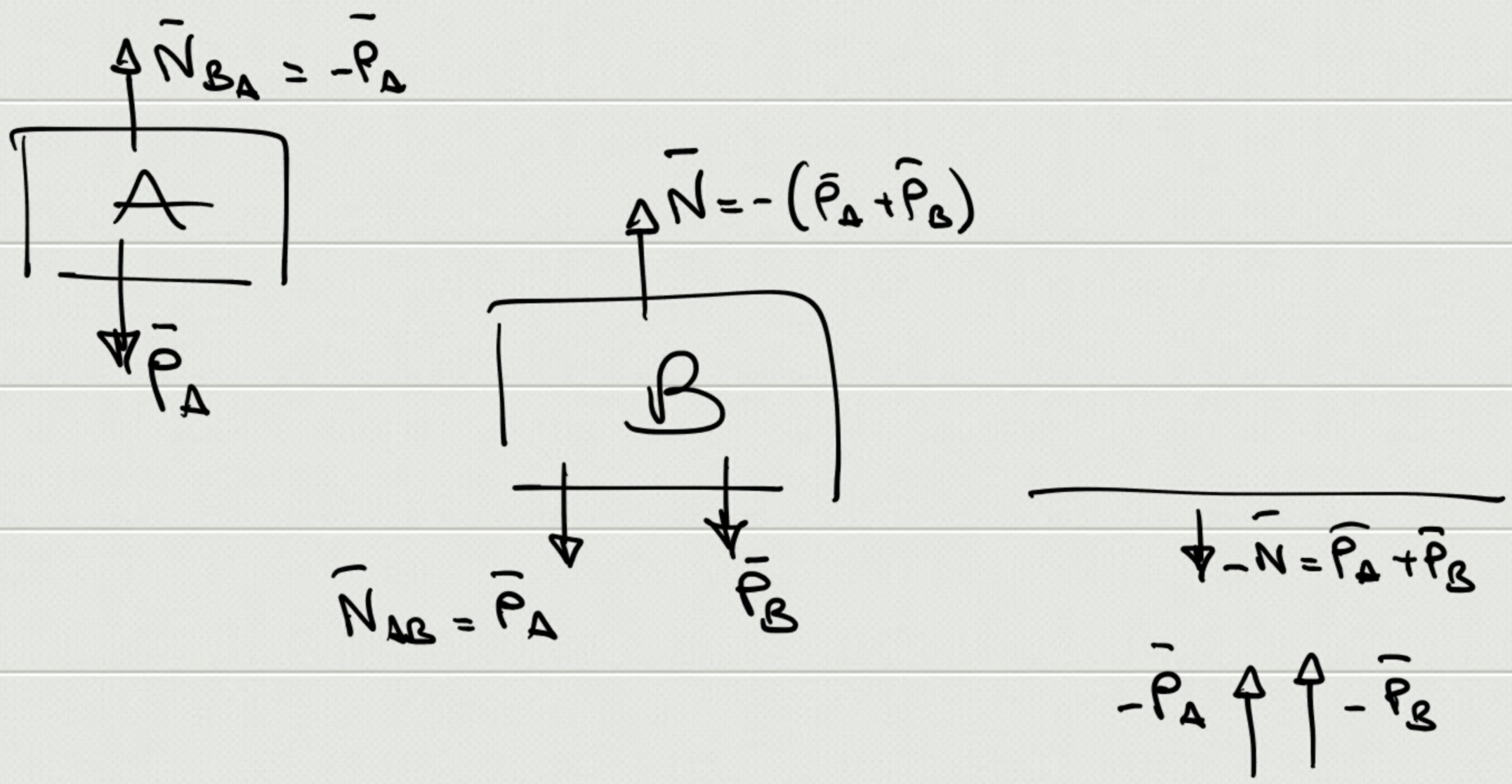


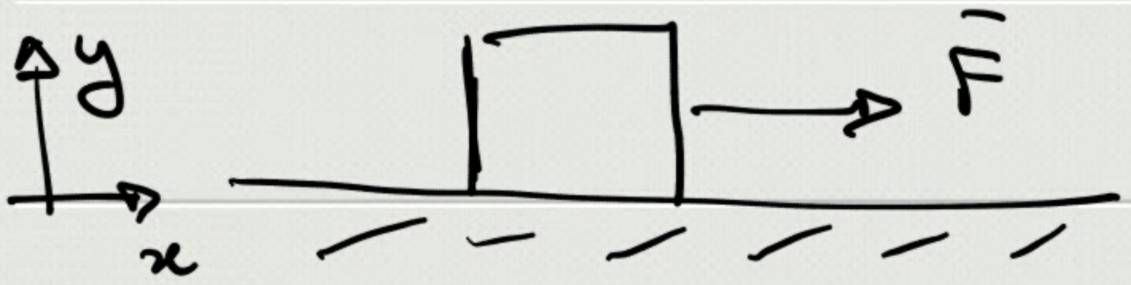







A
B


$\bar{N}_{BA} = -\bar{P}_A$
A
$\bar{P}_A$
$\bar{N} = -(\bar{P}_A + \bar{P}_B)$
B
$\bar{N}_{AB} = \bar{P}_A$
$\bar{P}_B$
$-\bar{N} = \bar{P}_A + \bar{P}_B$
$-\bar{P}_A$
$-\bar{P}_B$



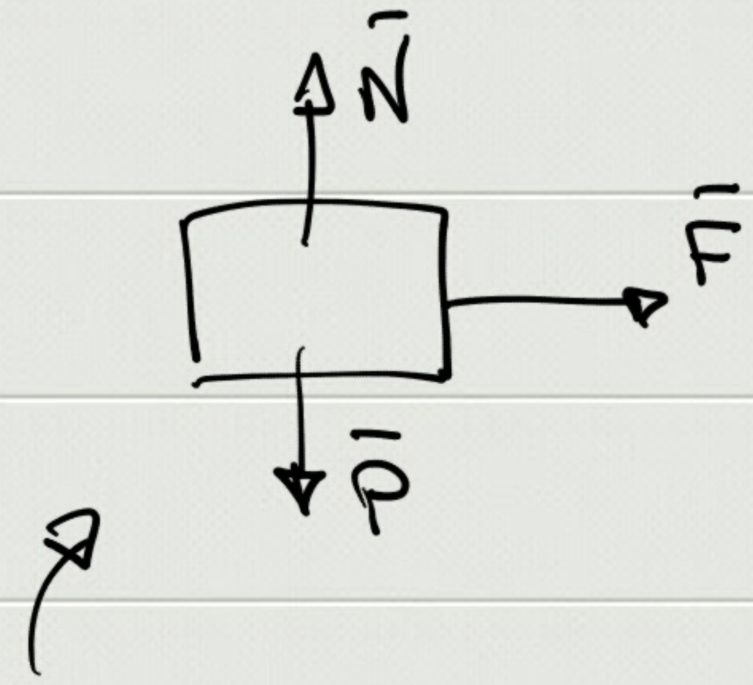


Diagramma di corpo libero

$$\sum_i \bar{F}_i = m\bar{a}$$

$$\bar{F} + \bar{P} + \bar{N} = m\bar{a}$$

$$\begin{cases} x: & F = m a_x \quad \Rightarrow \quad a_x = \dfrac{F}{m} \\ y: & N - P = m a_y = 0 \quad \Rightarrow \quad N = P \end{cases}$$

$$F = \text{cost} \quad \Rightarrow \quad a_x = \text{cost} \quad \Rightarrow \quad x(t) = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$$

$$x_0 = 0 \quad v_0 = 0 \quad \Rightarrow \quad \boxed{x(t) = \frac{1}{2}\left(\frac{F}{m}\right) t^2}$$